Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
Munshi Nagar, Andheri (W), Mumbai – 400 058.
(Autonomous Institute Affiliated to University of Mumbai)

3.4: Research Publications and Awards

3.4.3: Number of research papers published per teacher in the Journals as notified on UGC CARE list during the last five years

Research Journal Paper published in Year Jan-Dec 2021

	Research Paper Title	Author Names	Link to Paper
1	Cspo-Dcnn: Competitive Swarm Political Optimisation For Breast Cancer Classification Using Histopathological Image	Sudhir Dhage, Jyoti Kundale	https://www.tandfonline.com/doi/full/10.1080/21681163.2021.2012828
2	Multiclass Classification Of Histology Images Of Breast Cancer Using Improved Deep Learning Approach	Sudhir Dhage, Jyoti Kundale	https://link.springer.com/chapter/10.1007/978-981-16-3067-5_34
3	Comprehensive Survey Of User Behaviour Analysis On Social Networking Sites	Sudhir Dhage ,Pramod Bide	https://www.inderscience.com/offer.php?id=119601
4	Cross Domain-Based Ontology Construction Via Jaccard Semantic Similarity With Hybrid Optimization Model	Sudhir Dhage, Shital Kakad	https://www.citegraph.io/paper/60b9a409e4510cd7c8fa225f
5	Wavelet-Based Imagined Speech Classification Using Electroencephalography	Sudhir Dhage, Dipti Pawar	https://www.inderscience.com/info/inarticle.php?artid=121737
6	Detection Of Parkinson'S Disease Via A Multi-Modal Approach	Dr. Preetida Vinayakray-Jani,	https://doi.org/10.1109/ICCCNT51525.2021.9580135

		Dayanand Ambawade, Jinang Gandhi, Aumkar Gadekar, Tania Rajabally	
7	Automated Snake Bite Prevention System	Surekha Dholay, Kunal Jain, Shriya Khatri, Nishant Kabra	https://www.ijitee.org/wp-content/uploads/papers/v10i3/C83540110321.pdf
8	Cross Language Application For Disease Prediction	Kiran Gawande, Shivani Bhutala, Niket Parekh, Advait Lad	http://thegrenze.com/index.php?display=page&view=journalabstract&absid=725&id=8
9	E-Commerce Shopping Gets Real: The Rise Of Interactive Virtual Reality	Sunil Ghane, Satish R. Billewar, Karuna Jadhav, D. Henry Babu	https://bbrc.in/wp-content/uploads/2021/05/BBRC_Vol_14_No_05_Special-Issue_25.pdf
10	Eight shape Electromagnetic Band Gap structure for bandwidth improvement of wearable antenna	Vidya R. Keshwani, Pramod P. Bhavarthe and Surendra S. Rathod	https://www.jpier.org/PIERC/pier.php?paper=21070603
11	Tunable Triple Band-Notched UWB Antenna Using Single EBG and Varactor Diode	Vijay R. Kapure, Pramod P. Bhavarthe, and Surendra S. Rathod	https://www.jpier.org/PIERC/pier.php?paper=21012204
12	Applications of Artificial Intelligence in Fire and Safety	Surendra Rathod, Vedant Kumar and Siddhant Kumar	https://drive.google.com/file/d/1P_mDmHjOcj5HfXXp8cV-fgMmxgIEmK8N/view?usp=sharing
13	FPGA-Based Hybrid Control Strategy for Resonant Inverter in	Darshana Nimesh Sankhe ,	https://ieeexplore.ieee.org/document/9324745

	Induction Heating Applications"	Rajendra R. Sawant, and Y. Srinivasa Rao	
14	Hardware-in-the-Loop Simulation of Induction Heating System for Melting Applications Using Xilinx System Generator	D. N. Sankhe, R. R. Sawant and Y. S. Rao	https://link.springer.com/chapter/10.1007/978-981-15-8443-5_77
15	A Comparative Analysis based approach for Bitcoin Price Forecasting,	Yash Wadalkar, Yellamraju V H Sai Tarun, Jaiesh Singhal, Reena Sonkusare	https://www.ijert.org/research/a-comparative-analysis-based-approach-for-bitcoin-price-forecasting-IJERTV10IS040313.pdf
16	Dynamic Object Elimination based Indoor 3D Mapping System	Pritam Mane, Vedant Kumar, Sreekar L, Pallavi Malame, Reena Sonkusare, and Sukanya Kulkarni,	https://www.ijraset.com/fileserve.php?FID=32890
17	Classification Comparative Analysis for Detection of Brain Tumor Using Neural Network, Logistic Regression & KNN Classifier with VGG19 Convolution Neural Network Feature Extraction	Vijaya Kamble Dr. Rohin Daruwala	https://books.aijr.org/index.php/press/catalog/view/114/44/1480-1
18	An Integrated Method for 3D Reconstruction, Dimensional Analysis and Manipulation of an Object Based on Image Processing	Soham Pinge, Nitish Wadhavkar, Amey Singh, Amol Deshpande	https://www.ijraset.com/fileserve.php?FID=37614
19	Alert System of Earthquake Detection	Surya Prakash, Sujata Kulkarni	https://www.ijraset.com/fileserve.php?FID=37614
20	Counting repetitions using Sensor	Devang ,Taiyeeba , Dr.	https://drive.google.com/file/d/1pvTbUsowrETe1X3sCODVF05lkxDVCk8F/view

		Pooja Raundale, Harshil Kanakia	?usp=sharing
21	IoT-based water distribution monitoring system	Dr. Aarti Karande, Suyash Oza, Omkar Wadekar, Prachi Dalvi	https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3882439
22	Automated Dimension Measurement System	Shubham Kakirde, Shubham Jain, Swaraj Kaondal, Reena Sonkusare, Rita Das	https://www.ijeat.org/wp-content/uploads/papers/v10i5/D23990410421.pdf

Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization ›

Volume 10, 2022 - Issue 5

Research Article

CSPO-DCNN: Competitive Swarm Political Optimisation for Breast Cancer Classification using Histopathological Image

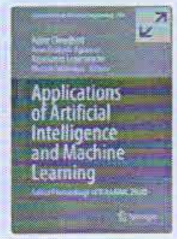
Jyoti Umesh Kundale & Sudhir Dhage

Pages 549-564 | Received 13 Mar 2021, Accepted 27 Nov 2021, Published online: 16 Dec 2021

Cite this article https://doi.org/10.1080/21681163.2021.2012828 Check for updates

ABSTRACT

Breast cancer is one of the most familiar diseases, and it is the second main reason with several risk stratification and subtypes in the woman. Furthermore, breast cancer is the second most important cause of women's death. In this paper, the Competitive Swarm Political Optimisation (CSPO)-based Deep Convolutional Neural Network (DCNN) is devised to classify breast cancer. The Type 2 Fuzzy and Cuckoo Search-based (T2FCS) filter is applied for the effective pre-processing process. In addition, the blood cell segmentation process is carried out using the colour-based thresholding model. The most important features, such as area, solidity, eccentricity, perimeter, entropy, kurtosis and Empirical Mode Decomposition (EMD),



Applications of Artificial Intelligence and Machine Learning pp 457–468

Multiclass Classification of Histology Images of Breast Cancer Using Improved Deep Learning Approach

Jyoti Kundale & Sudhir Dhage

Conference paper | First Online: 27 July 2021

Part of the Lecture Notes in Electrical Engineering book series (LNEE,volume 778)

Abstract

Breast Cancer has become a serious threat to women life in the overall word. It is necessary to diagnosis for Breast cancer early to avoid mortality rate. In the era of medical imagining with the advent of Artificial Intelligence systems, it has become easy to detect breast cancer at an early stage. Histopathology image modality is one of the best ways for breast cancer detection as it is easy to store in digital format for a long time. Breast cancer

INDERSCIENCE PUBLISHERS
Linking academia, business and industry through research

Comprehensive survey of user behaviour analysis on social networking sites

by Pramod Bide; Sudhir Dhage

International Journal of Computer Applications in Technology (IJCAT), Vol. 66, No. 1, 2021

Abstract: Social networking sites play an important role in every person's life. Users start expressing their emotions online whenever any humanitarian or crisis like event occurs. A lot of sub-events are stirred up and the internet gets flooded with people tweeting/posting their opinions. Identifying user behaviours, their content and their interaction with others can help in event prediction, cross event detection, user preferences, etc. For these reasons, our research process was divided into studying user behaviour with respect to content-centric, probabilistic approach and a hybrid incorporating the two. We further investigate the existence of multiple OSNs and how they affect user behaviour. The purpose of this paper is to investigate the existing research methodologies and techniques along with discussion and comparative studies. User behaviour analysis is carried out based on content centric, probabilistic and hybrid approach. Content centric analysis dealt with analysis of the content posted which gives rise to varied applications such as gender prediction, malicious users, real-time user preferences, emotional content influence on users etc. It is observed that in probabilistic approach, most of the papers addressed, employed the clustering mechanisms followed by probability distribution for the analysis of user behaviour.

Online publication date: Sat, 11-Dec-2021

Keep up-to-date
Our Blog
Follow us on Twitter
Visit us on Facebook
Our Newsletter (subscribe for free)
RSS Feeds
New issue alerts

ScienceDirect

Expert Systems with Applications
Volume 178, 15 September 2021, 115046

Cross domain-based ontology construction via Jaccard Semantic Similarity with hybrid optimization model

Shital Kakad, Sudhir Dhage

Show more

Share Cite

https://doi.org/10.1016/j.eswa.2021.115046
Get rights and content

Abstract

Semantic web technology seems to be in the infant stage as only little efforts have been taken on ontology construction with cross-domain application. This paper intends to take an effort on a new workspace, in which the ontology construction model under cross-domain application is performed. The core concern of this work is on two decision-making process namely data filtering and data annotation. Certain process is followed in this work: (i) Preprocessing (ii) Proposed Jaccard Similarity Evaluation (iii) Data filtering and Outlier Detection (iv) Semantic annotation and clustering. More particularly, data filtering is performed based on the evaluated similarity function. The outliers are identified and grouped separately. The data annotation is performed based on the semantics and thereby the clustering process takes place to form the ontology precisely. This clustering process obviously relies to the optimization crisis as the optimal centroid selection becomes the greatest issue. In order to solve this, this paper extends with the introduction of a hybrid algorithm named Circling Insisted-Rider Optimization Algorithm (CI-ROA), which hybrids the concept of Whale Optimization Algorithm (WOA) and Rider Optimization Algorithm (ROA), respectively. Finally, the performance of proposed work is compared and proved over other state-of-the-art models.

Introduction

Ontology (Navimipour et al., 2017, Nilashi et al., 2018, Awan et al., 2019, Wu et al., 2015) is considered as a normalized knowledge base that contains a set of concepts over a domain and the correlations of concepts. Ontologies offer the structural outlines to organize information and are deployed in various fields like semantic web, artificial intelligence, software engineering, systems engineering, library science, biomedical informatics, and enterprise bookmarking. In order to design the machine-understandable ontology (Qian et al., 2018) in different domain, the professionals derived the concept from system architecture. Ontology (Kermany and Alizadeh, 2017, Rivero et al., 2013, Liu et al., 2009) offers a support service for semantic-based heterogeneous resource service and search development via the normalized demonstration for domain relationships and concepts.

International Journal of Biomedical Engineering and Technology > 2022 Vol.38 No.3

Title: Wavelet-based imagined speech classification using electroencephalography

Authors: Dipti Pawar; Sudhir Dhage

Addresses: Department of Computer Engineering, Sardar Patel Institute of Technology, Andheri(W), Mumbai, India ' Department of Computer Engineering, Sardar Patel Institute of Technology, Andheri(W), Mumbai, India

Abstract: Oral communication is the natural way in which humans interact. However, in some circumstances, it is difficult to emit an intelligible acoustic signal, or it is desired to communicate without making sounds. In these conditions, systems that enable spoken communication in the absence of an acoustic signal is desirable. In this context, brain-computer interface (BCI) is a remarkable way of solving daily life problems. The major objective of the proposed research is to develop an imagined speech classification system based on electroencephalography (EEG); which consists of pre-processing, feature extraction and classification. In the pre-processing stage, EOG artefacts are removed via independent component analysis (ICA). Discrete wavelet transform (DWT) is used to extract wavelet-based features from EEG segments. Finally, the support vector machine (SVM) is employed for the discriminant of extracted features. This research achieves promising ends in classification accuracy compared with some of the most common classification techniques in BCI.

Keywords: electroencephalography; EEG; brain-computer interface; BCI; discrete wavelet transform; DWT; imagined speech; support vector machine; SVM.

DOI: 10.1504/IJBET.2022.121737

International Journal of Biomedical Engineering and Technology, 2022 Vol.38 No.3, pp.215 - 224

Received: 18 Oct 2018
Accepted: 16 Jan 2019
Published online: 07 Apr 2022

Access provided by:
Sardar Patel Institute of Technology

Sign Out

Detection of Parkinsons Disease Via a Multi-Modal Approach

Jinang Gandhi ; Aumkar Gadekar ; Tania Rajabally ; Preetida Vinayakray-Jani ; Dayanand Ambawade **All Authors**

Abstract: Parkinson's affects 700000 plus people in India. It primarily affects the nervous system. Its various symptoms manifest in the form of lack of motor skills, verbal and me... **View more**

Metadata

Abstract:
Parkinson's affects 700000 plus people in India. It primarily affects the nervous system. Its various symptoms manifest in the form of lack of motor skills, verbal and mental abilities, affected speech and memory. The disease has no cure and the best remedy is early diagnosis and treatment to reduce the severity of the symptoms. The lack of easy access to medical facilities and doctors is a major deterrent to early diagnosis of Parkinson's. Therefore, we propose here a solution in the form of a mobile application that allows people to test themselves for symptoms of Parkinson's anywhere, anytime. We use a two-way approach to detect the probability of the patient having Parkinson's. The authors posit image processing and machine learning techniques to detect anomalies in speech and tracing patterns. The model for the Trace test achieves a test accuracy of 97.39% (VGG-19)architecture and the Speech test achieves 93.2% (LGBM Classifier).

Published in: 2021 12th International Conference on Computing Communication and Networking Technologies (ICCCNT)

Date of Conference: 06-08 July 2021

INSPEC Accession Number: 21461544

Date Added to IEEE *Xplore*: 03 November 2021

DOI: 10.1109/ICCCNT51525.2021.9580135

ISBN Information:

Publisher: IEEE

OPEN ACCESS
International Journal of Innovative Technology and Exploring Engineering (IJITEE)
ISSN: 2278-3075 (Online), Volume-10 Issue-3, January 2021

Automated Snake Bite Prevention System

Kunal Jain, NishantKabra, Shriya Khatri, Surekha Dholay

Abstract—Snakebite isnt an illness without a treatment, anyway a substantial number of grievous setbacks go untreated each year. Many governments basically put this issue in the difficult to handle basket.Hence we have made a gadget which will recognize snakes using cameras and spurn the distinguished snake using a snake repeller.This should provoke an uncommon decrease in the number of snakebites and in like manner in the number of deaths.We are planning a basic, easy, convenient gadget which will automate the procedure of snake repeller.

Index Terms—Snake bite, Image processing, Edge detection, Gaussian filter, Sobel filter

I. INTRODUCTION

As per Government of India information, there were 61,507 snake bites with deaths of 1124 during 2006; 76,948 bites and 1359 deaths in 2007.In April, 2015 Union Minister of Health, Family Welfare M.Nadda said that around 1,124 individuals had died in 2012 because of snakebites while 1,008 had died in 2013. People, for the most part poor farming workers, die from snakebites consistently - deaths that can be evaded if the issue got its due from general well being policymakers. The Nadsur and Dhondse towns of the Raigad locale are two of the numerous towns confronting this deadly issue because of snakes. There was an absence of efficient readiness to battle this issue.Hence to determine this issue we are building up a simple device which will have the capacity to recognize snakes and alert nearby prople about the snakes.

II. RELATED WORK

Generally, snake repellers come in pellet or spray styles. It appears, in view of purchaser, that pellets are considerably more viable outside than sprays as is evident from [1]. The way the pellet and spray style deterrents work is by interfering with the snakes feeling of smell. This neurological interruption makes them abandon the territory, leaving to discover 'natural air' to chase prey. For open air utilize, purchasers say the Ortho Snake B Gon is the best choice. Produced using basic oils, it's showcased as a naturally well disposed alternative that won't hurt people, pets or plants. It's intended to keeps snakes from entering the home, and also settling and searching.

Revised Manuscript Received on January 08, 2021.
* Correspondence Author
Kunal Jain*, Sardar Patel Institute of Technology, Mumbai, India. E-mail.id. jainkunal31@gmail.com
Nishant Kabra, Sardar Patel Institute of Technology, Mumbai, India. E-mail.id. nishantkabra123@gmail.com
Shriya Khatri, Sardar Patel Institute of Technology, Mumbai, E-mail.id. khatrishriya@gmail.com
Surekha Dholay, Sardar Patel Institute of Technology, Mumbai, India. E-mail.id. surekha dholay@spit.ac.in

You should simply sprinkle it around your property each 30 days. There are various gadgets accessible globally as "Home and Garden" snake and rat repellent gadgets. These gadgets deal with the standard of using sonic wave sound and vibrations [5].

As the sonic wave is produced while going through a media like soil, it likewise makes vibrations because of energy transfer. In spite of the fact that snakes can not hear sound waves they can detect vibrations, that is somewhat hearing for them. The basic premises of these devices are to create stimuli to prevent chances of conflict by reducing the animals desire to enter or stay in the area where these vibrations are happening. In the gadget that has been spread like an out of control fire via web-based networking media as the " Farmer's magic stick", the above principle and technology has been innovated into portable devices that couldve been based on the traditional method of Hitting a stick on the ground while walking in the dark so that if a snake is in your way, it will sense the vibration and move away. The old principle works because of its very short term window of avoiding snake in front of you when you are walking, the target objective is to warn the snake of your presence expecting it to move away from your path and not to run meters away.

Fig. 1. Ortho Snake Repellent

There are various gadgets accessible globally as "Home and Garden" snake and rat repellent gadgets. These gadgets deal with the standard of using sonic wave sound and vibrations [5]. As the sonic wave is produced while going through a media like soil, it likewise makes vibrations because of energy transfer. In spite of the fact that snakes can not hear sound waves they can detect vibrations, that is somewhat hearing for them. The basic premises of these devices are to create stimuli to prevent chances of conflict by reducing the animals desire to enter or stay in the area where these vibrations are happening. In the gadget that has been spread like an out of control fire via web-based networking media as the " Farmer's magic stick", the above principle and technology has been innovated into portable devices that couldve been based on the traditional method of Hitting a stick on the ground while walking in the dark so that if a snake is in your way, it will sense the vibration and move away. The old principle works because of its very short term window of avoiding snake in front of you when you are walking, the target objective is to warn the snake of your presence expecting it to move away from your path and not to run meters away.

search this site

Home About Us Journals Conferences Indexing Digital Library Search Contact Us

Cross Language Application for Disease Prediction

Journal: GRENZE International Journal of Engineering and Technology

Authors: Shivani Butala, Advait Lad, Niket Parekh, Kiran Gawande

Volume: 6 **Issue:** 2

Grenze ID: 01.GIJET.6.2.10 **Pages:** 114-122

Abstract

The infrequent and inadequate medical facilities for the people residing in the remote villages of India due to the lack of medical experts is a glaring medical issue facing the country. The proposed cross language application aims to solve the aforementioned problem with the use of technology by utilizing Translation techniques and Machine Learning models. With the usage of the proposed application, the users can vocally input their experienced symptoms into the application in their local language and get as output, the disease they possibly might be suffering from within seconds. To increase the accuracy of the results, an ensemble approach is followed in the paper which is a combination of five Machine Learning Models and a majority decision technique is applied to get the most accurate results. The proposed application is able to predict the disease with an accuracy of 90% in return of the symptoms provided by the user.

<< BACK

GIJET

TITLE:
GRENZE International Journal of Engineering and Technology

EDITOR IN CHIEF:
Dr. Janahanlal Stephen (Professor, Computer Science & Engg. Dept, Viswajyothi College of Engineering &Technology. Affiliated to KTU University, Kerala, S.India.)

ISSN:
2395-5295(Online); 2395-5287(Print)

FREQUENCY:
Twice Yearly

ASSOCIATED PUBLISHER:
GRENZE

RELATED SUBJECTS:
Computer Science & Engineering, Electrical & Electronics Engineering, Mechanical Engineering, Civil Engineering

RESEARCH GROUP:
GRENZE Engineering Group

INDEXING:

OFFICIAL WEBSITE:
http://gijet.thegrenze.com/

Biosc.Biotech.Res.Comm. Special Issue Vol 14 No 05 (2021) Pp-130-132

BBRC
Bioscience Biotechnology Research Communications

E-Commerce Shopping Gets Real: The Rise of Interactive Virtual Reality

Satish R. Billewar[1], Karuna Jadhav[2], D. Henry Babu[3] and [4]Sunil Ghane
[1]*Vivekanand Institute of Management Studies & Research, Mumbai, India*
[2]*Neville Wadia Institute of Management Studies & Research, Pune, India*
[3]*St. Francis Institute of Management and Research, Mumbai, India*
[4]*Sardar Patel Institute of Technology, Mumbai, India*

ABSTRACT

Virtual Reality is considered to be the technology that will be used very extensively. The paper discusses the idea of the implementation of virtual reality in E-commerce. The shopper can virtually walk through the aisles of the store, stop and examine an item. The photos, textual information, support this examination process, and audio clips that describe the item. Depending on the photos' completeness, the shopper can virtually touch and examine the merchandise from all angles, and the order can be placed into the shopping cart. The paper is an effort to give the customer a more real-world shopping experience without going outside. It also makes shopping online more attractive and reliable. The user can have the store experience by looking for the items from the isles and have a better experience in evaluating the items by having the ability to pick them up like in actual stores.

KEY WORDS: VIRTUAL REALITY, E-COMMERCE, VIRTUAL STORE, CUSTOMER SATISFACTION, SHOPPING CART.

INTRODUCTION

The virtual store's experience is also enhanced by an augmented reality assistant that helps the users by giving them all the information needed in audio form or using its avatar. One primary advantage of having a virtual store is its possibility of the plugin the store in different games targeting the virtual video game players. They could be more interested in the idea of buying products even while playing their favorite game and could order their merchandise from an e-commerce store within the game. The satisfaction from the virtual immersion is enormous as everything can be customized according to the user, like crowd, products, and interaction satisfaction during shopping can be given without any hassle.

Biosc Biotech Res Comm P-ISSN: 0974-6455 E-ISSN: 2321-4007
Corresponding author email:

Identifiers and Pagination
Year: 2021 Vol: 14 No (5) Special Issue
Pages: 130-132

Article Information
Received: 20th Jan 2021
Accepted after revision: 30th Mar 2021

DOI: *http://dx.doi.org/10.21786/bbrc/14.5/25*

2. Concept Discussion: Virtual reality has been one of the emerging technologies that have existed for a long time as a concept. From medieval panoramic paintings to Morton Heilig'sSensorama patented in 1962, the point of virtual reality is to trick someone's brain into believing something is real, even when it isn't. Virtual Reality application has always been in some kind of simulation like flight simulation or in movie watching. These both have existed from the early 1900s, with the first flight simulator built-in 1929 by Edward Link and Sensorama (1962), which was used for movie watching.

3. Existing System: The existing system is mostly browser-based with functionalities like Superpose an item from the shopping cart into the Audio/Visual Virtual Tour. So that the shopper can "see' that item on the shelf of the store. The shopper can "pick up that item, examine it by viewing additional images and textual descriptions and pricing data, and even asking a clerk a specific question. Finally, if desired, place the item into their shopping cart.

Figure.1 shows the software stack of the virtual reality shopping software components and will be referenced when discussing the main functions of the virtual reality

Progress In Electromagnetics Research C, Vol. 116, 37–49, 2021

Eight Shape Electromagnetic Band Gap Structure for Bandwidth Improvement of Wearable Antenna

Vidya R. Keshwani[1], Pramod P. Bhavarthe[2], and Surendra S. Rathod[1, *]

Abstract—In this paper, a rectangular eight shaped Electromagnetic Band Gap (EBG) structure at 5.8 GHz Industrial, Scientific and Medical (ISM) band for wearable application is proposed with intent to improve the impedance bandwidth of antenna. The unit cell of an EBG structure is formed using eight shape on outer ring with inner square patches. The simulation of the eight shape EBG unit cell is carried out using eigen mode solution of Ansys High Frequency Structure Simulator (HFSS). Simulated results are validated by experimental results. The application of proposed EBG for an inverse E-shape monopole antenna at 5.8 GHz is also demonstrated. Band stop property of EBG structure reduces surface waves, and therefore, the back lobe of a wearable antenna is reduced. The frequency detuning of antenna takes place due to high losses in human body. Suitably designed EBG structure reduces this undesirable effect and also improves front to back ratio. The proposed compact antenna with designed EBG has observed the impedance bandwidth of 5.60 GHz to 6.15 GHz which covers 5.8 GHz ISM band. Evaluation of antenna performance under bending condition and on-body condition is carried out. Effectiveness of EBG array structure for Specific Absorption Rate (SAR) reduction on three layer body model is demonstrated by simulations. Calculated values of SAR for tissue in 1 g and 10 g are both less than the limitations. In conclusion, it is appropriate to use the proposed antenna in wearable applications.

1. INTRODUCTION

A wearable antenna is an antenna that is specially designed to function while being worn on the body. These antennas can be integrated into clothing. Wearable antennas for in and on Body Area Networks (BAN) are designed by textile materials. It can communicate with other antennas on body surface or with an external antenna. These antennas are used in Wireless Body Area Networks (WBAN) for applications like health care, military, sports, etc. [1]. Due to such a wide application scope, in recent past new antenna topologies for wearable application including multiband, compact and wide band antennas have been rapidly evolving.

The general performance requirements for wearable antennas are low mutual influence between antenna and human body for high antenna efficiency, low SAR, small size, low profile, and Improved bandwidth [2]. Large bandwidth of textile antennas is needed as it makes them less sensitive to detuning arising from the adverse environmental influences. There are many methods to increase the bandwidth of textile antennas. SAR is a parameter used to evaluate power absorption in human tissue. Electromagnetic power absorbed by the body may pose potential health risks. Hence, the evaluation of SAR is also an important consideration in wearable antenna design. SAR should be well below standard acceptable limit. Many organizations including IEEE, ICNIRP, FCC have recommended limits

Received 6 July 2021, Accepted 24 September 2021, Scheduled 10 October 2021
* Corresponding author: Surendra Singh Rathod (surendra_rathod@spit.ac.in).
[1] Department of Electronics Engineering, Sardar Patel Institute of Technology, Andheri(W), Mumbai 400058, India. [2] Department of Electronics and Telecommunications Engineering, Padmabhushan Vasantdada Patil Pratisthan College of Engineering, Mumbai 400022, India.

Progress In Electromagnetics Research C, Vol. 110, 181–195, 2021

Tunable Triple Band-Notched UWB Antenna Using Single EBG and Varactor Diode

Vijay R. Kapure[1, *], Pramod P. Bhavarthe[2], and Surendra S. Rathod[3]

Abstract—In this paper, a UWB monopole antenna with triple band-notch characteristics using single TBMV-EBG (Triple band multi-via electromagnetic bandgap) unit cell is proposed and demonstrated. The antenna with a fork-type radiating patch with TBMV-EBG is simulated using Ansys HFSS. Measurement results show triple band-notches at 3.39, 5.78, and 8.60 GHz, respectively, which are in good agreement with simulation results. The proposed antenna has bi-directional pattern in E-plane and omnidirectional pattern in H-plane. Moreover, tunable characteristics of the proposed antenna using a single varactor diode are also presented. By changing the capacitance of varactor, the band-notched antenna is effectively tuned from 2.69–3.46, 5.71–7.84, and 8.40–8.50 GHz. The same antenna structure can be operated at different band notching modes depending upon the varactor's capacitance. Therefore, the proposed UWB antenna will prove to be a promising candidate wherein multi-band rejections using single TBMV-EBG unit cell and tunability using one varactor diode are desirable.

1. INTRODUCTION

Ultra-wideband (UWB) systems have become important for short range, high data rate, low power consumption, and indoor data communication. In 2002, Federal Communications Commission(FCC) allocated the frequency band of 3.1 to 10.6 GHz for UWB systems. In this allocated UWB, other narrowband systems like wireless local area network (WLAN) for IEEE 802.11a operating at (5.15–5.35) and (5.725–5.825) GHz & WIMAX (3.3–3.6) GHz and X-band (8–12) GHz are already present. Therefore, in order to avoid electromagnetic interference, UWB antenna should reject these types of narrowband systems. In recent years, many techniques have been proposed to reject these narrow bands. In [1–4], narrow band notches are achieved by etching slots on radiating elements. In [5, 6], quad and quintuple UWB band-notched antennas are reported using CSRRs and RCSRRs, respectively. However, this technique of inferring the antenna to get multi-bands affects the performance of antenna. To overcome this problem, placing a parasitic element near the feed line of antenna is proposed in [7–16]. In [7], a hexagon-shaped monopole UWB antenna with two same size edge located via (ELV) EBG cells are placed near the feed line to get dual band notches at 3.5 GHz and 5.5 GHz. In [8], a UWB circular monopole antenna with dual band notch rejection is achieved using four center located via (CLV) EBGs. In [9], two modified mushroom type semicircular EBG structures with dissimilar sizes are placed near the feed line to produce dual band notch rejection in WIMAX band. In [10], to get triple band notch characteristics, three EBG unit cells are used. In [11], a UWB monopole antenna with two spiral EBGs, one on front side near the feed and the other on back side, is used to produce dual band notches at the upper and lower WLAN frequencies. In [12], an antenna is reported with

Received 22 January 2021, Accepted 23 February 2021, Scheduled 27 February 2021
* Corresponding author: Vijay Ramesh Kapure (vijaykapure777@gmail.com).
[1] Department of Electronics Engineering, Sardar Patel Institute of Technology, Andheri (West), Mumbai-400058, India. [2] Department of Electronics and Telecommunication Engineering, Padmabhushan Vasantdada Patil Pratishthan's College of Engineering, Sion, Mumbai-400022, India. [3] Department of Electronics Engineering, Sardar Patel Institute of Technology, Andheri (West), Mumbai-400058, India.

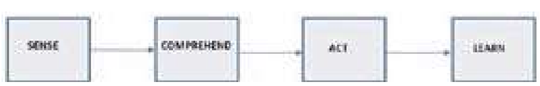
APPLICATIONS OF ARTIFICIAL INTELLIGENCE IN FIRE & SAFETY

- Student Chapter Contribution from S.P.I.T Mumbai

Overview of the applications of AI in mitigating the dangers due to fire

In this article, we'll look at how techniques like computer vision, machine learning, and deep learning can help the front line workers to make more informed decisions in a stressful situation. The data coming from the sensors would have a positive impact—both in terms of rescue time and the life safety of the victim as well as the fire respondent.

What is AI?



Any system that utilizes AI possesses the following 4 stages:

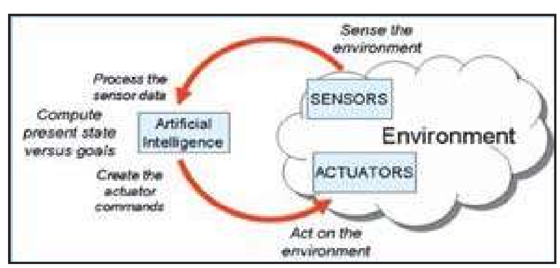
1. Sense. AI lets a machine perceive the world around it by acquiring and processing images, sounds, speech, text, and other data. Sensing of data can take place through various sensors available in the surrounding or by utilizing the data available on the internet.

2. Comprehend. AI enables a machine to recognize patterns and context in the information it collects, just as humans interpret information by understanding patterns and context in their perception of the world. This stage also includes the cleaning of the acquired data for a better understanding.

3. Act. AI allows the machine to take actions in the physical or digital world based on that perception or comprehension.

4. Learn. AI enables a machine to continuously optimize its performance by learning from the success or failure of its actions. This helps in obtaining better accuracy overtime of the machine.

The figure below depicts the process involved while applying AI into the real environment.



AI for front line workers

A common problem faced by the fire officials when a rescue operation is underway in a place that has been engulfed with flames and smoke is poor visibility.

1-21

FPGA-Based Hybrid Control Strategy for Resonant Inverter in Induction Heating Applications

Darshana Nimesh Sankhe, Rajendra R. Sawant, *Senior Member, IEEE*, and Y. Srinivasa Rao, *Member, IEEE*

***Abstract*—An implementation of a fully digital control system for driving resonant power converter is a challenging task. Furthermore, solving discrete-time equations on a digital platform like field-programmable gate array (FPGA) device being used for its fast response characteristic along with the real-time validation is an additional prominent challenge. This article aims at solving both the challenges for a series resonant inverter used in solid-state induction heating (IH) application. The proposed hybrid control strategy is aimed at phase-shifted pulsewidth modulation to achieve output power regulation and pulse frequency modulation control to maintain soft switching for dynamic load conditions, simultaneously. The performance of the controller is investigated under varying load and power conditions. An improvement in the overall control performance is ensured by introducing a dynamic slope compensation logic with digital blocks. The simulation-based modeling of control system and proportional–integral–differential tuning methods are presented. Furthermore, the proposed controller is validated by hardware-in-the-loop simulation for FPGA implementation and tested experimentally on a laboratory prototype of 5-kW IH system for metal melting applications, using Zynq XC7Z020-1clg484 FPGA platform.**

***Index Terms*—Discrete-time control, field-programmable gate array (FPGA), hardware-in-the-loop (HIL), induction heating (IH), resonant inverter.**

I. Introduction

RECENTLY, an exponential growth in microelectronic devices and power semiconductor technology has stimulated the development of controllers for power converters using reliable digital platforms. Traditionally, digital controllers like microprocessor, microcontroller, and digital signal processors (DSP) are used to develop and implement control algorithms for power converters. These platforms often execute different control functions sequentially, which are not suited for fast control algorithms and parallel signal processing [1]. The field-programmable gate array (FPGA) based control system designs are in various stages of research and are in great demand because of their inherent parallel processing capabilities. The FPGA's low latency, quick response time together with reconfigurable and reprogrammable characteristics are best suited for real-time controller implementation as well as rapid prototyping of application-specific integrated circuits (ASIC), for power control applications [2], [3]. However, it is a well-known fact that developing hardware-description language (HDL) code for real-time control application is a challenging task, thus the system development time increases substantially with the increase in complexity of control algorithm. In order to reduce the development time, a model-based design approach with Xilinx system generator (XSG) is being used, which provides a virtual FPGA environment and supports hardware-in-the-loop (HIL) simulation [4].

The electromagnetic induction heating (IH) technique has been practiced for domestic, medical, and industrial applications over other heating methods due to its high performance, energy efficiency, smooth power control, safety and environmental cleanliness [5]. The high-frequency resonant inverter forms an important building block of a solid-state IH system. Traditionally, in resonant inverters, the output power has been controlled by varying the switching frequency, which ensures zero voltage switching (ZVS) when operated above resonance frequency [6], [7]. However, the resonant inverter has a problem of controlling electromagnetic interference and it needs complex filtering, exhibits low efficiency due to high-frequency operation mainly in low-medium output power regions, and insufficient use of inductive components [6], [8]. These drawbacks can be overcome by using fixed-frequency power control methods such as pulse density modulation (PDM), phase shift control (PSC), and asymmetrical-duty cycle or asymmetrical pulsewidth modulation (PWM) control. The PDM technique can provide a wide output power range but lacks precise control with high efficiency and subjected to discontinuous output current with flicker emission problems, especially at the light-load conditions [9].

The PSC technique regulates the power by varying the phase of the switching control pulses and results in a symmetrical voltage cancellation. The asymmetrical PWM technique makes use of uneven duty-cycle operation of the switching devices and controls the output power. Among these fixed-frequency control methods, PSC is the preferred strategy for resonant inverters due to its controller simplicity, ability to provide symmetric pulses, ability to provide soft switching with no additional circuitry, and elimination of preregulation stage. These fixed-frequency control techniques lose ZVS when resonant inverters are operating over a wider power range [10], [11]. Thus, to maintain ZVS at

Manuscript received May 6, 2020; revised September 8, 2020 and November 24, 2020; accepted December 23, 2020. Date of publication January 14, 2021; date of current version December 23, 2021. *(Corresponding author: Darshana Nimesh Sankhe.)*

Darshana Nimesh Sankhe is with the Department of Electronics and Telecommunication Engineering, Sardar Patel Institute of Technology, Mumbai 400058, India, and also with the Dwarkadas Jivanlal Sanghvi College of Engineering, Mumbai 400056, India (e-mail: darshanasankhe78@gmail.com).

Rajendra R. Sawant and Y. Srinivasa Rao are with the Department of Electronics and Telecommunication Engineering, Sardar Patel Institute of Technology, Mumbai 400058, India (e-mail: rajendra.sawant@spit.ac.in; ysrao@spit.ac.in).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/JESTIE.2021.3051584.

Digital Object Identifier 10.1109/JESTIE.2021.3051584

2-2021

Hardware-in-the-Loop Simulation of Induction Heating System for Melting Applications Using Xilinx System Generator

Darshana N. Sankhe, Rajendra R. Sawant, and Y. Srinivasa Rao

Abstract Hardware-in-the-loop (HIL) simulation is increasingly used in development and testing of industrial systems and many more applications. This paper presents HIL simulation of induction heating (IH) system used for metal melting applications, wherein a resonant inverter with IH load and control logic, both, are modeled using high-level Xilinx System Generator (XSG). A phase shift control technique with continuous resonant frequency tracking mechanism is deployed for power regulation, which improves the efficiency of the IH system. Performance analysis of the complete system is carried out using HIL simulation, and results are validated for varying power conditions. Hardware resource utilization, timing, and power requirements are analyzed by real-time implementation on Zynq XC7Z020-1clg484 FPGA board. This complete system modeling with the HIL simulation approach allows through testing of the system with different control algorithms before the commencement of physical systems, which further reduces risk of device damage, controller development time, initial cost and supports rapid prototyping of control strategy on high-speed FPGA platforms.

Keywords Hardware-in-the-loop · Digital controller · FPGA · Induction heating · Xilinx System Generator

D. N. Sankhe (✉) · R. R. Sawant · Y. S. Rao
Sardar Patel Institute of Technology, Mumbai, India
e-mail: darshanasankhe78@gmail.com

R. R. Sawant
e-mail: rajendra.sawant@spit.ac.in

Y. S. Rao
e-mail: ysrao@spit.ac.in

A. P. Pandian et al. (eds.), *Proceedings of International Conference on Intelligent Computing, Information and Control Systems*, Advances in Intelligent Systems and Computing 1272, https://doi.org/10.1007/978-981-15-8443-5_77

3-21

Published by :
http://www.ijert.org

International Journal of Engineering Research & Technology (IJERT)
ISSN: 2278-0181
Vol. 10 Issue 04, April-2021

A Comparative Analysis based approach for Bitcoin Price Forecasting

Yash Wadalkar, Yellamraju V H Sai Tarun, Jaiesh Singhal
UG Student
Dept. of Electronics and Telecommunication
Sardar Patel Institute of Technology
Mumbai, India

Prof. Reena Sonkusare
Head of Department
Dept. of Electronics and Telecommunication
Sardar Patel Institute of Technology
Mumbai, India

***Abstract*— Bitcoin, one of the most famous and high-in- demand cryptocurrencies, is a type of digital asset that is extremely difficult to track and make predictions upon. In addition, Bitcoin price does not correlate with market-movements, therefore, predicting its price action and its locus is an ordeal. In this paper, we have followed a comparative analysis approach, wherein we are using four different models to predict the trend of BTC Time series data. The results justify that the models have achieved accurate forecasting trends. During the period of 16th to 31st December 2020, Bitcoin prices experienced considerably high swings, due to the increased demand for it. In quantitative terms, the prices experienced fluctuations to the tune of 8000 USD. Despite these enormous price changes, we were able to achieve a model, that helped us attain a Mean Absolute Error (MAE) of 153.55 USD and Mean Square Error (MSE) of 43231.80 USD. Conventional Bitcoin price predicting researches follow a single to two model approach. However, for a highly volatile asset like Bitcoin, making long-term predictions and generalizing them based on limited number of models results in low accuracy outputs. This gap has been bridged in our research, we have worked with different models, as well as fragmented the time intervals into smaller portions, post which the prediction was made for only 2 days. Using this approach, we attained results with least error rates. The results obtained clearly show that ARIMA is the best model for predicting the future trends for BTC time series data. It takes into account the different types of decompositions like Regular Trend, Sessional and Residual Trend making the model give the best results.**

Keywords—Ccryptocurrency; blockchain; bitcoin; Time-series Analysis; Facebook Prophet; LSTM; ARIMA; XGBoost

I. INTRODUCTION

Technology has been disrupting the way humans live, work and even transact. Globally, economies and financial institutions have been going digital at an unprecedentedly fastpace[1]. The advent of Fin-Tech has left the conventional financial system archaic. One of the most recent developments in this space is the advent of Cryptocurrencies, more specifically, the most talked about, Bitcoin. Bitcoin, one of theoriginal and first cryptocurrencies, currently has a market capitalization north of around 600 billion USD, which would only grow steeply in the forthcomingyears.

This digital currency derives its popularity from two features: its decentralized nature [2] and the extraordinary volatility [3] that this asset exhibits. The returns that the investors yield on this investment is significantly higher than that of traditional banking or market schemes. Moreover, since there is no government intervention, the price is solely controlled by the public, for the public.

Generalising, the prices of cryptocurrencies are not driven by the market therefore technical analysis becomes important in determining the price range of bitcoin over a period. Since technical analysis does not depend on external economic data, solely past price patterns are used to predict the price of Bitcoin [4].

In recent researches, LSTM and ARIMA models were used to forecast prices of Indian stocks for a period of 5 months. LSTM showed better results compared to ARIMA [5]. Another research where Gradient Boost Tree Model was used to capture Twitter data and public sentiments for a tenure of 3.5 weeks, analysed each tweet, for which over 50 percent accuracy was achieved [6]. A study carried out by Wint forecasted the closing price of the Myanmar Stock Price Index (MYANPIX) using ARIMA and Facebook Prophet. For all three periods (daily, weekly and monthly) prophet, having less error rate, has outperformed ARIMA [7].

Understanding the recent developments and with reference to researches published in this domain, we have carried out condensation of various approaches, in our research, which were followed to make predictions for equity markets and other cryptocurrencies. A comparative analysis of four models has been incorporated to furnish the best possible prediction model, wherein each model individually, contributes to the increased efficiency of the entire mechanism.

II. METHODOLOGY

The below drawn flowgraph shows our approach towards building the models and comparing them based on various performancemetrics.

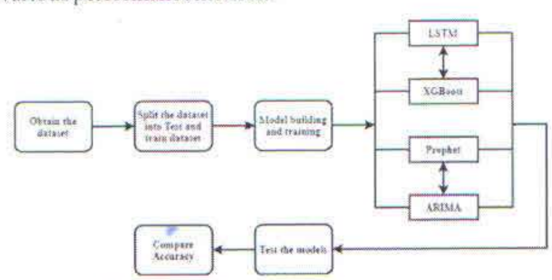


Fig 1. Block Diagram of the proposed approach

International Journal for Research in Applied Science & Engineering Technology (IJRASET)
ISSN: 2321-9653; IC Value: 45.98; SJ Impact Factor: 7.429
Volume 9 Issue I Jan 2021- Available at www.ijraset.com

Dynamic Object Elimination based Indoor 3D Mapping System

Pritam Mane[1], Vedant Kumar[2], Sreekar L[3], Pallavi Malame[4], Reena Sonkusare[5], Sukanya Kulkarni[6]
[1, 2, 3, 4, 5, 6]*Department of Electronics and Telecommunication, Bharatiya Vidya Bhavans Sardar Patel Institute of Technology*

Abstract: In today's world, the need for autonomous robots is increasing at an exponential rate and the implementation of Simultaneous Localisation And Mapping (SLAM) is gaining more and more attention. One of the major component of SLAM is 3D Mapping of the environment which enables autonomous robots to perceive the environment like a human does for which many Depth cameras or RGB-D cameras prove useful. This paper proposes a continuous real-time 3D mapping system that tackles the long existing problem of point cloud distortion induced by dynamic objects in the frame. Our method uses the Microsoft Kinect V1 as the RGB-D camera and the packages in the Robotic Operating System (ROS) like the Real Time Appearance Based Mapping (RTAB-map) for 3D reconstruction. A ROS based method is used to implement dynamic object elimination in real-time. For the purpose of dynamic objects detection in the frame, two algorithms - Deep Learning based tiny YOLO-v3 and a Machine Learning based Haar Cascade classifier are used. The results from the two are compared in terms of accuracy, execution time and mean Average Precision (mAP) and it was inferred that although Haar Cascade model is comparatively less accurate when detecting objects, it is two times faster than YOLO which makes the system more real-time. The real-time implementation was given more preference while selecting the model.
Keywords: Simultaneous Localization and Mapping(SLAM), 3D mapping, Robot Operating System (ROS), Deep learning, Stereo Vision, Object Detection, Point Cloud, Scale invariant feature transform (SIFT)

I. INTRODUCTION

In today's world where the need of autonomous robots is ever increasing, the implementation of Simultaneous Localisation And Mapping (SLAM) is gaining more and more attention. Robotic cleaners and inventory management robots are autonomous and have the same functionality in both the cases, that is, to navigate and find their way through an unknown environment. In order to do so, they need to have a frame of reference and perceive the nearby environment (through computer vision or through Lidar sensors) to sketch a map of its surroundings and traverse its path. 3D reconstruction involves using computer vision to perceive the external surroundings an building a depth based 3D point cloud. Hence, the underlying principle is a common intersection point between 3D reconstruction and autonomous navigation. Autonomous robot navigation is a step ahead of 3D reconstruction. The rise in demand for 3D solutions calls for a holistic solution that can perceive the surroundings around and create a 3D map of the same.

The usage of computer vision for robotic automation and programming robots or creating 3D layouts has been quite prevalent. The SLAM (simultaneous localization and mapping) problem has persisted for a long time and a lot of research done portrays new ways and methodologies to tackle the problem for mapping. Robot navigation has long been an important and active area of research.

Existing innovations in this field utilize expensive cameras for creating depth and disparity maps which tend to be accurate, but the overall cost of the product increases. Other techniques include using stereo vision cameras to calculate the depth which can further be used to create 3D point clouds. This method of reconstruction is fairly cheaper but accuracy is compensated. The existing systems work very well for reconstruction of static frames but when dynamic objects enter the frame the output is not very accurate. Our system proposes a method for detecting and removing the dynamic objects from the frame in order to create an accurate 3D reconstruction of a scene.

The pipeline for the 3D reconstruction consists of 4 major steps:

1) Preparing the camera system used for retrieving image and depth data
2) Creation of depth and disparity maps
3) Generating and aligning point clouds
4) Refining and processing the point clouds

Active research has been done in all the domains. The methods for 3D reconstruction and point cloud generation using stereo vision or depth estimation using Lidars have been implemented. Novel methods for depth estimation using Convolutional Neural networks have been previously performed. A work we referred, describes a unique depth sensing approach using convolutional neural networks (CNN)[1]. This falls under monocular SLAM vision which uses only a single camera, maybe coupled with other sensors for depth estimation. Although this eliminates the need of RGB-D cameras the process gives high errors.

5-2021

Classification Comparative Analysis for Detection of Brain Tumor Using Neural Network, Logistic Regression & KNN Classifier with VGG19 Convolution Neural Network Feature Extraction

Vijaya Kamble*[1], Dr. Rohin Daruwala[2]

[1]Sardar Patel Institute of Technology, Andheri West
[2]Veermata Jijabai Technological Institute Matunga Mumbai

*Corresponding author
doi: https://doi.org/10.21467/proceedings.114.6

Abstract

In recent years due to advancements in digital imaging machine learning techniques are used in medical image analysis for the prognosis and diagnosis of various abnormalities in the human body. Various Machine learning algorithms, convolution and deep neural networks are used for classification, detection and prediction of various brain tumors. The proposed approach is a different comparative classification analysis approach which is based on three different classification namely KNN classifier,Logistic regression & neural network as classifier. It is based on a deep learning feature extraction technique using VGG19. This VGG 19-layer image recognition model trained on Imgenet. Generally, MRI data sequences are analyzed in terms of different modalities and every modality contains rich tissue information. So, feature exaction from MRI sequences is very important task for brain tumor classification. Our approach demonstrated fair classification on BRATS Benchmarks 2018 data set with different modalities and sizes of images,results are without any human annotations. Based on selected classifiers all the classifiers gives accuracy above 90%. It is good compared to other state of art methods.

Keywords: MRI, images, Brain tumor segmentation, KNN, Neural network Logistic regression machine learning, VGG19.

1 Introduction

Most challeging medical field now a days using automated machine learning based classification of various types body images like CT Scan MRI Ultrasound images for accurate study of abnormalities ,cancers & cysts dignosis. In this type of studies researchers and professionals take help of various types of classifiers and these classifiers employs different types of features like morphological features such as area,height, width perimeter and many more.Most life threatning disease is brain tumor. Brain tumours are abnormal masses in or on the brain. It is uncontrolled cell proliferation growth of a cell as tumour .It may occur as a result of, a failure of the normal pattern of cell death, or both. Brain tumours can be either primary or secondary. Each of these tumours has unique biological , radiographical, and clinical characteristics that dictate, in part, their management. Malignant tumours generally grow fast and can spread to tissues in close proximity. Malignancy mostly refered to cancer. Benign tumours do not spread and grow slowly Eventhough benign tumours are serious and can be life threatening, growing in a confined limited space, a benign tumour can exert pressure on the brain and compromise its function. The location of the tumour is key to diagnosis. Just use of MRI cannot reliably differentiate between the different types of tumours on the basis of imaging.The information of tumour location can be very helpful for prediction of the exact histology of the tumours .

MRI has been accepted as the major diagnostic modality for the brain tumor analysis. Comparing with CT scan MRI are more sensitive for brain tumours. MRI used for both for detection of tumours as well as in showing more completely the extent of the tumour spread either residual or recurrent disease.

Proceedings DOI: 10.21467/proceedings.114; Series: AIJR Proceedings; ISSN: 2582-3922; ISBN: 978-81-947843-8-8

6-2021

International Journal of Engineering and Advanced Technology (IJEAT)
ISSN: 2249-8958 (Online), Volume-10 Issue-4, April 2021

An Integrated Method for 3D Reconstruction, Dimensional Analysis and Manipulation of an Object Based on Image Processing

Soham Pinge, Nitish Wadhavkar, Amey Singh, Amol Deshpande

Abstract: ***Analysis of dimensions of a product, its visualization as a 3D model & volume estimation, positively impact the operations of manufacturing department in an industry. In this paper, a system has been designed to integrate the proposed features like 3D reconstruction, dimensional analysis & support for manipulating the reconstructed model. Research was carried out to estimate the volume of objects using their virtual 3D model & allow the user to edit this model which later can be compared with the original object, for manipulation. The pictures of an object from different planes are captured and using image processing, it is reconstructed in 3D, which is used to estimate parameters like volume, surface area, maximum dimensions across different planes. This edited 3D model will be utilized for generating a binary file representing the changes. This system provides detailed information to the user along with flexibility, thus enhancing the way in which the manufacturing industry works, as it requires no manual input or interaction once the process begins.***

Keywords: ***Industrial Solutions, Mathematics and Computing, Image Processing, Volume Estimation, 3D Reconstruction, Dimensional Analysis, Edge Detection.***

I. INTRODUCTION

In manufacturing industry, analysis of dimensions of objects is of utmost importance. And this should be carried out as efficiently and safely as possible, considering production for sensitive domains where non-destructive processes should be used. Also, with the COVID-19 pandemic, manufacturing and testing has become a huge challenge for industries with paucity of staff, further increasing the need for automation at every level. The demand for non-contact based industrial solutions has grown exponentially.

Manuscript received on April 20, 2021.
Revised Manuscript received on April 26, 2021.
Manuscript published on April 30, 2021.

* Correspondence Author

Soham Pinge*, Department of Electronics & Telecommunications, Sardar Patel Institute of Technology, Mumbai, India. Email: soham.pinge@spit.ac.in

Nitish Wadhavkar, Department of Electronics & Telecommunications, Sardar Patel Institute of Technology, Mumbai, India. Email: nitish.wadhavkar@spit.ac.in

Amey Singh, Department of Electronics & Telecommunications, Sardar Patel Institute of Technology, Mumbai, India. Email: amey.singh@spit.ac.in

Dr. Amol Deshpande, Department of Electronics & Telecommunications, Sardar Patel Institute of Technology, Mumbai, India. Email: amol_deshpande@spit.ac.in

In today's age, automation is of prime importance. Operations like 3D Reconstruction and volume estimation are performed regularly in industries, but manually and separately. Automation will save effort and the time invested in manual execution. This project not only introduces methods for automation of the said processes but also consolidates these using step by step into codependent procedures for integrating them with each other. Therefore, this paper proposes a methodology for the improvement of the quality of industrial automation, manufacturing and testing, that uses new techniques and integrates the processes of dimensional analysis, volume estimation, 3D reconstruction and provides support for manipulation of an object, which can then be exported in the form of a standard .stl file format to be utilized as preferred by the user.

II. LITERATURE SURVEY

Before diving deep into the paper, we shall look at some related works. There are two main techniques for 3D reconstruction, LIDAR and photogrammetry. LIDAR uses laser to scan objects and are more accurate and convenient. But, in design and manufacturing industries, the real time images of objects are necessary as operations like logo marking, designing, colouring, etc. are involved. In such applications, photogrammetry is a suitable technique. In [1] only orthogonal projections have been considered, therefore no 3d reconstruction has been performed. In [2] we see that deep learning has been implemented, which requires presence of large databases, providing a hindrance in our case because there is no existence of a useful dataset and even if there was, its storage would have further used up a lot of memory. Also, the use of a single camera is time consuming, has a high margin of error and maybe require additional complex robotics. In [3] although machine learning is not used, the method of Scale Invariant Feature Transform (SIFT) is used. This method too requires a significant number of key-points stored in a database, giving rise to the same earlier mentioned problems. In [4] a method known as Structure from motions is used. It mainly describes parameters such as effective position and attribute calculation of the camera and our object. The smoothing of the images is not obtained upto a great ex- tent. In [5] a volume measurement system for calculating volume of tomatoes is proposed. This provides a good and fast solution, but it comes at the cost of accuracy which reaches a maximum of 80% despite capturing five different images.

50

International Journal for Research in Applied Science & Engineering Technology (IJRASET)
ISSN: 2321-9653; IC Value: 45.98; SJ Impact Factor: 7.429
Volume 9 Issue VIII Aug 2021- Available at www.ijraset.com

Alert System of Earthquake Detection

Surya Prakash[1], Sujata Kulkarni[2]
[1]*Department of Electronics And Telecommunication, Bharatiya VidyabhavansSardar Patel Institute Of Technology, Mumbai*
[2]*Associate Professor, Bharatiya Vidyabhavans Sardar Patel Institute of Technology, Mumbai*

Abstract: Global warning shows unpredictablenature that changes in subsurface geologicfeatures. Due to the complex nature ofseismic events, it is challengeable task to efficiently identify the prominent features that leads to seismic events. Taking the advantage of availability of Seismic dataset, AI using machine learning is a powerful statistical tools to mitigate these practical challenges for earthquake prediction. The paper focuses on the alert and prediction model of an earthquake using machine learning algorithm. The alert time is a function of distance from epicenter and most alert time is for high magnitude earthquake, because high magnitude earthquake rupture over much larger area and take time to propagate, thus delayed for warning. The main aim of the project is to build precise earthquake prediction model using XGboost regression in machine learning technique. The XGboost regression model implemented on the dataset of 14700 data records. The data is gathered from the Kaggle platform that contains data on events of the earthquake in Indian Subcontinent of 18 years and the last updated is of the year 2018. Tableau 2019 is explored for data visualization of the predicted outcome. The proposed technique is able to give precise prediction of earthquake with sufficient time that will reduce the maximum damages and many lives will be saved.

Keywords: Seismic signals, XGboost, Tableau, Prediction.

I. INTRODUCTION

Earthquake is floor shaking caused by a sudden motion of rock in the earth's crust. Such moves occur alongside faults, which can be skinny zones of overwhelmed rock setting apart blocks of crust. Earthquake prediction research has been conducted for more than 100 years withoutapparent success. Allegations of the breaches failed to withstand scrutiny. Extensive searches failed to find reliable precursors. The theoreticalwork suggests that error is a non-linear process that is highly sensitive to minute details that cannot be measured for the state of the Earth in large volume, and not just in the immediate vicinity of the hypocenter. Thus any small earthquake has some probability of graduating into a large event. It appears that reliable warnings of impending large earthquakes arevirtually impossible.

Algorithms have been proposed for earthquake prediction using expert system. The main objective of the proposed framework is to identify and evaluate methods, models, parameter and equipment used to forecast earthquakes using different features present in the dataset. The aim is to explore different tools to improve the accuracy / precision of forecast model. It is quite difficult to build a precisemodel because many factors are related for the occurrence of earthquake, like depth, temperature, magnitude etc. The main focus isto provide accurate prediction of earthquake event using machine learning approach. The proposed prediction model can give precise prediction of earthquake event with sufficient time that will help the government to take the necessary measures to prevent maximum damages and save maximum lives. XGboost is a powerful machine learning method which can research extraordinarily complex features.

II. LITERATURE SURVEY

Frequency filtering is extensively used in the habitual processing of seismic records to enhance the signal-to-noise ratio (SNR) of the recorded indicators and as a consequence to enhance subsequent analyzes, primarily based on a deep neural network.[1]The performance ofthe Deep Denoiser was impressive even after it did not result in perfect signal-to-noiseseparation. [2]Subsequent results showed when using wireless sensor network (WSN) , smart phones could be used not only as a recording tool, but also as a highly accurate seismic detection tool. Early warning of 5 seconds was achieved. [3]Internet of things (IOT) is a network of computed physical objects that enables the connection, collection and exchange of data in the conditions of the physical environment. Early warning system implementation via lab view software, Alert signal is sent to smart phones and is carried out by IOT. The hardware part helps to detect the signal , the software part is used to deliver the warning signal to the human. [4] The developed early warning system can trigger a local protection coverage (alarms, disconnectionof hazardous gadget) to reduce the effect of a seismic episode. [5]Another implementationwas tested in the WSN high-speed rail earlywarning system. The problem in the WSN is thelack of technology to provide information on surface movement characteristics, spectrum or time series, and to trigger an alarm to minimize casualties. [6]

International Journal for Research in Applied Science & Engineering Technology (IJRASET)
ISSN: 2321-9653; IC Value: 45.98; SJ Impact Factor: 7.429
Volume 9 Issue VI Jun 2021- Available at www.ijraset.com

Counting Repetitions using Sensor

Dr. Pooja Raundale[1], Prof. Harshil Kanakia[2], Devang Chhajed[3], Taiyeeba Chikhalia[4]
[1]H.O.D. Masters of Computer Application, Sardar Patel Institute of Technology, Mumbai, India
[2, 3, 4]Master's of Computer Application, Sardar Patel Institute of Technology, Mumbai, India

Abstract: This paper describes the step by step process for counting the number of repetition using different Mobile Phone Sensors. The Algorithm is tested on various datasets acquired by the people. To keep it simple and accessible to low-end devices, the algorithm needed to be simple and fast. We identify the most volatile axis, and then smooth the curve to remove any noise, 1 cycle being the 1 complete motion we calculate the number of complete cycles by the axis wave which gives us the total number of repetitive motions completed.
Keywords: sensors, motion sensors; accelerometer; gyroscope; linear acceleration; gravity; magnetometer; mobile phone sensor

I. INTRODUCTION

In recent years, the world has seen a boom in fitness bands like mi band, fitbit, amaze bit, etc. which ranges from Rs. 2000 to Rs. 6000. Its hardware is fixed and cannot be updated[1]. Today everyone owns a smartphone, which is a very powerful device bundled with various types of sensors for measuring the acceleration, rotation, gravity, proximity, etc. With the boom in the smart wearables market, they have become an integral part of people's life. Our paper aims to leverage the sensors in them to detect the repetitive motion of the user[2]. This data can further be used for analysis.

II. LITERATURE SURVEY

There are various ways to count the repetitive motions which are currently being used which are based on Dynamic Time Warping(DTW)[3], neural network-based detection[4], etc[5][6][7]. Most of them require either a machine learning algorithm or some part of machine learning algorithm which require a lot of processing power.
Most of the current methods to count repetitive motion compares two waves which occur consecutively to detect motion. It needs an activity(motion) to compare the user's current activity with, which requires a user to pre-select the activity which they want to track.

III. THE PROCESS

A. Data Collection

We developed an application to log the user movement and asked different people to perform various activities. We recorded data of 3 main sensors i.e. accelerometer, gyroscope, rotation sensor [8][9][10].

B. Re-align all the axis to start from the same point

We use various sensors such as accelerometer, gyroscope, rotation sensor, etc to capture movements. But as we can see in fig1 below, all three axes start from a different position. Hence deciding which axis to work on becomes a little harder since every axis is starting from a different point. Therefore, we need to get all the axes to start from the same point, i.e. 0.

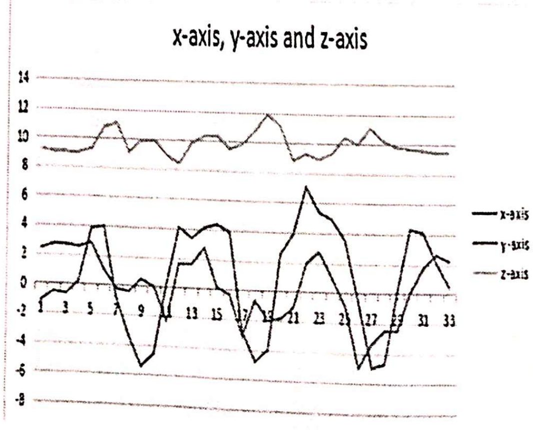


Fig 1: Raw Data from Accelerometer for moving left-right 3 times.

AY2021
AK

IoT-based water distribution monitoring system

Proceedings of the International Conference on IoT Based Control Networks & Intelligent Systems - ICICNIS 2021

5 Pages
Posted: 8 Jul 2021 ·

Prachi Dalvi (https://papers.ssrn.com/sol3/cf_dev/AbsByAuth.cfm?per_id=4762290)
Master of Computer Applications, Sardar Patel Institute of Technology, Andheri

Aarti Karande (https://papers.ssrn.com/sol3/cf_dev/AbsByAuth.cfm?per_id=4762292)
Master of Computer Applications, Sardar Patel Institute of Technology, Andheri

Suyash Santosh Ojha (https://papers.ssrn.com/sol3/cf_dev/AbsByAuth.cfm?per_id=4762294)
Master of Computer Applications, Sardar Patel Institute of Technology, Andheri

Omkar Ravindra Wadekar (https://papers.ssrn.com/sol3/cf_dev/AbsByAuth.cfm?per_id=4762299)
Master of Computer Applications, Sardar Patel Institute of Technology, Andheri

Date Written: July 8, 2021

Abstract

Water is precious natural resource for sustaining life and environment. Effective and sustainable management of water resources is vital for ensuring sustainable development. In this paper, we analyze the water problems in India and find an efficient solution to automate and monitor the overall process. Water scarcity is one of the major constraints in producing food, maintaining health, ecosystem protection, and social security. Increasing competition and conflicts pose social and ecological risks when it comes to water management [2]. Along with scarcity the conflicts are also likely to grow within sectors.[3] A smart city is a portal to a new age. Many houses exist today in which people can have amenities according to their comfort. Usage of the water is based on the people's population, season, etc, So, water supply management is required along with its maintenance. In methods being used today, many of us may not know how much water is used or wasted. To ensure the automation of the system as well as to avoid the wastage of the water, proposed system has been developed for smooth water supply management with the help of IoT devices.[4]

Download This Paper (Delivery.cfm/SSRN_ID3882439_code4548898.pdf?abstractid=3882439&mirid=1)

Open PDF in Browser (Delivery.cfm/SSRN_ID3882439_code4548898.pdf?abstractid=3882439&mirid=1&type=2)

22 References

1. Water Resources Scenario of India: Emerging Challenges and Management Options
Posted: 2007-03

2. Water Management Problems And Challenges In India: An Analytical Review (M. Dinesh Kumar
Posted: 2000-01
Crossref (https://doi.org/10.1201/9780429316234)

3. Water Management Problems And Challenges In India: An Analytical Review (M. Dinesh Kumar
Posted: 2000-01
Crossref (https://doi.org/10.1201/9780429316234)

4. R Priya, G P Rameshkumar
A Novel Method to Smart City's Water Management System With Sensor Devices and Arduino
International Journal of Computational Intelligence Research

Automated Dimension Measurement System

Shubham Kakirde, Shubham Jain, Swaraj Kaondal, Reena Sonkusare, Rita Das

Abstract: ***In this fast-paced world, it is inevitable that the manual labor employed in industries will be replaced by their automated counterparts. There are a number of existing solutions which deal with object dimensions estimation but only a few of them are suitable for deployment in the industry. The reason being the trade-off between the cost, time for processing, accuracy and system complexity. The proposed system aims to automate the mentioned tasks with the help of a single camera and a line laser module for each conveyor belt setup using laser triangulation method to measure the height and edge detection algorithm for measuring the length and breadth of the object. The minimal use of equipment makes the system simple, power and time efficient. The proposed system has an average error of around 3% in the dimension estimation.***

Keywords: Laser, Warehouses, Dimension Measurement, Automation.

I. INTRODUCTION

Efficient warehouse management is critical for an effective supply chain in the world where e-shopping is getting popular [1], [2]. A lot of warehouses even systems today employ manual labor for monotonous jobs like measuring dimensions of objects, scanning codes, etc. These systems are not only slow, but also not as cost effective as their automated counterparts [3], [4]. Also, in a post-Covid world, contact-less delivery will play a major role in warehouse systems, ensuring the safety of the customers. Many E-Commerce and E-Grocer Companies have their sales increased by 100% due to Covid19 [5]. There has been a drastic increase in the sales of packed foods, sanitizers and other household things. According to a report an E-Grocer company received 30,000 orders per day due to high requirement. Thus, as the demand rises, compromise in the services takes place. This may include wrong delivery of a product, delay in the shipment. Thus, a system is required that is faster, is contact less and involves fewer manual efforts. 3D reconstruction of objects using 2D images have been a great advancement in the field of computer vision. This can be done to analyze the object without any physical intervention in the process which makes it suitable for fragile objects. 3D reconstruction of objects also makes it easier for computational analysis and extracting the attributes of the object [6]–[8]. There are several solutions to solve the mentioned problem of estimating the dimensions of boxes. Each method employs a different set of hardware and different algorithmic process to achieve the same. Some focus extensively on the accuracy of the measurement while some focus on the operational speed and computational time. The power consumption varies based on the hardware employed. Height of a 3-D object can be measured with the help of a line laser and a camera. This is done with the help of laser triangulation method. The deflection in the line laser is checked by an algorithm to determine the shape of the surface of the object. Thus, this model works correctly for objects of any shape since the top surface is considered. Since, there is no estimation of the length and breadth of the objects, this model is not well suited for industrial applications where all the three dimensions of the object are required. The system that we are designing will consist of a single line laser and a camera module. This system focuses on utilizing as low power as possible because of a smaller number of components and high speed of operation which would be favorable for most of the industrial applications. The trade-off between the accuracy and the speed of operation is well suited for industrial needs. From algorithmic perspective, the only major operations would be masking the unwanted sections of the image and detecting the red laser line. This makes the process faster than the other observed methods which involve multiple lasers and cameras.

II. METHODOLOGY

In order to compute the three dimensions (length, breadth and the height) of an object, the initial step has to be calibrating the camera to map the pixel values to the real-world dimensions (in centimeters). This can be done by placing a line (of known length) of a distinct color on the ground level and detecting the number of pixels it takes in the image, this can be seen in the figure 5. To identify the line on the surface we use edge detection algorithm [9]–[11]. The length and breadth that the camera will see depends upon the height of the box under the camera (closer objects appear larger). Hence, we first need to compute the height of the box. The displacement in the line laser projected on the box is directly proportional to the height of the box as shown in figure 4.

Once the displacement is obtained, the height of the box can be calculated. The camera being a point camera, there will be error introduced in the height of the box as shown in figure 2.

Manuscript received on April 02, 2021.
Revised Manuscript received on June 07, 2021.
Manuscript published on June 30, 2021.
* Correspondence Author

Shubham Kakirde*, Department of Electronics & Telecommunications, Sardar Patel Institute of Technology, Mumbai (MH), India. Email: shubham.kakirde@spit.ac.in

Shubham Jain, Department of Electronics & Telecommunications, Sardar Patel Institute of Technology, Mumbai (MH), India. Email: shubham.jain@spit.ac.in

Swaraj Kaondal, Department of Electronics & Telecommunications, Sardar Patel Institute of Technology, Mumbai (MH), India. Email: swaraj.kaondal@spit.ac.in

Dr.Reena Kumbhare, Department of Electronic & Telecommunications Sardar Patel Institute of Technology, Mumbai (MH), India. Email: reena kumbhare@spit.ac.in

Dr. Rita Das, Department of Electronics & Telecommunications, Sardar Patel Institute of Technology, Mumbai (MH), India. Email: rita das@spit.ac.in